
Academic Positions

March 2020 to present **Assistant Professor**, *Stanford University*

Education

August 2011 to February 2020 **Ph.D., Computer Science**, *Massachusetts Institute of Technology*

August 2009 to August 2011 **M.S. Computer Science**, *University of Illinois at Urbana-Champaign*

August 2002 to June 2005 **B.E., Computer Science**, *Norwegian University of Science and Technology in Gjøvik*

Publications

Conference and Journal Publications

- PLDI 26 A Compiler for Fused Relational Operations on Multisets. James Dong and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue PLDI*, 2026
- PLDI 26 Bonsai: Compiling Queries to Pruned Tree Traversals. Alexander J Root, Christophe Gyurgyik, Purvi Goel, Kayvon Fatahalian, Jonathan Ragan-Kelley, Andrew Adams, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue PLDI (Distinguished Paper Award)*, 2026
- PLDI 26 Decoupling Data Layouts from Bounding Volume Hierarchies. Christophe Gyurgyik, Alexander J Root, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue PLDI*, 2026
- OSDI 26 Optimal Software Pipelining and Warp Specialization for Tensor Core GPUs. Rupanshu Soi, Rohan Yadav, Fredrik Kjolstad, Alex Aiken, Maryam Dehnavi, Michael Garland, and Michael Bauer. *Proceedings of the Symposium on Operating System Design and Implementation*, 2026
- OOPSLA 26 Deegen: A JIT-Capable VM Generator for Dynamic Languages. Haoran Xu and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 10, Issue OOPSLA*, 2026
- ASPLOS 26 FuseFlow: A Fusion-Centric Compilation Framework for Sparse Deep Learning on Streaming Dataflow. Rubens Lacouture, Nathan Zhang, Ritvik Sharma, Marco Siracusa, Fredrik Kjolstad, Kunle Olukotun, and Olivia Hsu. *International Conference on Architectural Support for Programming Languages and Operating Systems*, 2026

- CGO 26 Fast Autoscheduling for Sparse ML Frameworks. Bobby Yan, Alexander J Root, Trevor Gale, David Broman, and Fredrik Kjolstad. *International Symposium on Code Generation and Optimization*, 2026
- CGO 26 Ember: A Compiler for Embedding Operations on Decoupled Access-Execute Architectures. Marco Siracusa, Olivia Hsu, Victor Soria-Pardos, Joshua Randall, Arnaud Grasset, Eric Biscondi, Doug Joseph, Randy Allen, Fredrik Kjolstad, Miquel Moretó Planas, and Adrià Armejach. *International Symposium on Code Generation and Optimization*, 2026
- OOPSLA 25 REPTILE: Performant Tiling of Recurrences. Muhammad Usman Tariq, Shiv Sundram, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 9, Issue OOPSLA*, 2025
- MICRO 25 A Probabilistic Perspective on Tiling Sparse Tensor Algebra. Ritvik Sharma, Zi Xue, Nathan Zhang, Rubens Lacouture, Fredrik Kjolstad, Sara Achour, and Mark Horowitz. *Proceedings of the 58th IEEE/ACM International Symposium on Microarchitecture*, 2025
- JSSC 25 Onyx: A 12-nm Programmable Accelerator for Dense and Sparse Applications. Kalhan Koul, Olivia Hsu, Yuchen Mei, Sai Gautham Ravipati, Maxwell Strange, Jackson Melchert, Alex Carsello, Taeyoung Kong, Po-Han Chen, Huifeng Ke, Keyi Zhang, Qiaoyi Liu, Gedeon Nyengele, Zhouhua Xie, Akhilesh Balasingam, Jayashree Adivarahan, Ritvik Sharma, Christopher Torng, Joel Emer, Fredrik Kjolstad, Mark Horowitz, and Priyanka Raina. *IEEE Journal of Solid-State Circuits*, 2025
- IEEE Micro 25 Designing Programmable Accelerators for Sparse Tensor Algebra. Kalhan Koul, Zhouhua Xie, Maxwell Strange, Sai Gautham Ravipati, Bo Wun Cheng, Olivia Hsu, Po-Han Chen, Mark Horowitz, Fredrik Kjolstad, and Priyanka Raina. *IEEE Micro*, 2025
- ASPLOS 25 Automatic Tracing in Task-Based Runtime Systems. Rohan Yadav, Michael Bauer, David Broman, Michael Garland, Alex Aiken, and Fredrik Kjolstad. *Architectural Support for Programming Languages and Operating Systems*, 2025
- ASPLOS 25 Composing Distributed Computations Through Task and Kernel Fusion. Rohan Yadav, Shiv Sundram, Wonchan Lee, Michael Garland, Michael Bauer, Alex Aiken, and Fredrik Kjolstad. *Architectural Support for Programming Languages and Operating Systems*, 2025
- CGO 25 Stardust: Compiling Sparse Tensor Algebra to a Reconfigurable Dataflow Architecture. Olivia Hsu, Alexander Rucker, Tian Zhao, Varun Desai, Kunle Olukotun, and Fredrik Kjolstad. *International Symposium on Code Generation and Optimization*, 2025
- OOPSLA 24 Compiling Recurrences over Dense and Sparse Arrays. Shiv Sundram, Muhammad Usman Tariq, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 8, Issue OOPSLA*, 2024
- OOPSLA 24 Compilation of Shape Operators on Sparse Arrays. Alexander J. Root, Bobby Yan, Peiming Liu, Christophe Gyurgyik, Aart J.C Bik, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 8, Issue OOPSLA*, 2024
- OOPSLA 24 Compiler Support for Sparse Tensor Convolutions. Peiming Liu, Alexander J. Root, Anlun Xu, Yinying Li, Fredrik Kjolstad, and Aart J.C Bik. *Proceedings of the ACM on Programming Languages, Volume 8, Issue OOPSLA*, 2024
- ISCA 24 The Dataflow Abstract Machine Simulator Framework. Nathan Zhang, Rubens Lacouture, Gina Sohn, Paul Mure, Qizheng Zhang, Fredrik Kjolstad, and Kunle Olukotun. *Proceedings of the International Symposium on Computer Architecture (Distinguished Artifact Award)*, 2024

- PLDI 24 Compilation of Modular and General Sparse Workspaces. Genghan Zhang, Olivia Hsu, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 8, Issue PLDI*, 2024
- VLSI 24 Onyx: A 12nm 756 GOPS/W Coarse-Grained Reconfigurable Array for Accelerating Dense and Sparse Applications. Kalhan Koul, Maxwell Strange, Jackson Melchert, Alex Carsello, Yuchen Mei, Olivia Hsu, Taeyoung Kong, Po-Han Chen, Huifeng Ke, Keyi Zhang, Qiaoyi Liu, Gedeon Nyengele, Akhilesh Balasingam, Jayashree Adivarahan, Ritvik Sharma, Zhouhua Xie, Christopher Torng, Joel Emer, Fredrik Kjolstad, Mark Horowitz, Priyanka Raina. *IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, 2024
- HPCA 24 Revet: A Language and Compiler for Dataflow Threads. Alexander Rucker, Shiv Sundram, Coleman Smith, Matt Vilim, Raghu Prabhakar, Fredrik Kjolstad, and Kunle Olukotun. *IEEE International Symposium on High-Performance Computer Architecture*, 2024
- SC 23 Legate Sparse: Distributed Sparse Computing in Python. Rohan Yadav, Wonchan Lee, Melih Elibol, Manolis Papadakis, Taylor Lee-Patti, Michael Garland, Alex Aiken, Fredrik Kjolstad, and Michael Bauer. *ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis*, 2023
- PLDI 23 Indexed Streams: A Formal Intermediate Representation for the Fused Execution of Contraction Operations. Scott Kovach, Praneeth Kolichala, Timothy Guo, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 7, Issue PLDI*, 2023
- PLDI 23 Mosaic: An Interoperable Compiler for Tensor Algebra. Manya Bansal, Olivia Hsu, Kunle Olukotun, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 7, Issue PLDI (Distinguished Paper Award)*, 2023
- ASPLOS 23 BaCO: A Fast and Portable Bayesian Compiler Optimization Framework. Erik Hellsten, Artur Souza, Johannes Lenfers, Rubens Lacouture, Olivia Hsu, Adel Ejje, Fredrik Kjolstad, Michel Steuwer, Kunle Olukotun, and Luigi Nardi. *International Conference on Architectural Support for Programming Languages and Operating Systems*, 2023
- ASPLOS 23 The Sparse Abstract Machine. Olivia Hsu, Maxwell Strange, Ritvik Sharma, Jaeyeon Won, Kunle Olukotun, Joel Emer, Mark Horowitz, and Fredrik Kjolstad. *Architectural Support for Programming Languages and Operating Systems*, 2023
- TACO 23 Unified Buffer: Compiling Image Processing and Machine Learning Applications to Push-Memory Accelerators. Qiaoyi Liu, Jeff Setter, Dillon Huff, Maxwell Strange, Kathleen Feng, Mark Horowitz, Priyanka Raina, and Fredrik Kjolstad. *ACM Transactions on Architecture and Code Optimization (TACO)*, 2023
- TECS 23 Aha: An agile approach to the design of coarse-grained reconfigurable accelerators and compilers. Kalhan Koul, Jackson Melchert, Kavya Sreedhar, Leonard Truong, Gedeon Nyengele, Keyi Zhang, Qiaoyi Liu, Jeff Setter, Po-Han Chen, Yuchen Mei, Maxwell Strange, Ross Daly, Caleb Donovick, Alex Carsello, Taeyoung Kong, Kathleen Feng, Dillon Huff, Ankita Nayak, Raj Setaluri, James Thomas, Nikhil Bhagdikar, David Durst, Zachary Myers, Nestan Tsiskaridze, Stephen Richardson, Rick Bahr, Kayvon Fatahalian, Pat Hanrahan, Clark Barrett, Mark Horowitz, Christopher Torng, Fredrik Kjolstad, and Priyanka Raina. *ACM Transactions on Embedded Computing Systems*, 2023
- CGO 23 Looplets: A Language For Structured Coiteration. Willow Ahrens, Daniel Donenfeld, Fredrik Kjolstad, and Saman Amarasinghe. *International Symposium on Code Generation and Optimization*, 2023

- SC 22 SpDISTAL: Compiling Distributed Sparse Tensor Computations. Rohan Yadav, Alex Aiken, and Fredrik Kjolstad. *ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis*, 2022
- TACO 22 Compiler Support for Sparse Tensor Computations in MLIR. Aart J.C. Bik, Penporn Koanantakool, Tatiana Shpeisman, Nicolas Vasilache, Bixia Zheng, and Fredrik Kjolstad. *ACM Transactions on Architecture and Code Optimization*, 2022
- PLDI 22 DISTAL: The Distributed Tensor Algebra Compiler. Rohan Yadav, Alex Aiken, and Fredrik Kjolstad. *ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2022
- PLDI 22 Autoscheduling for Sparse Tensor Algebra with an Asymptotic Cost Model. Peter Ahrens, Fredrik Kjolstad, and Saman Amarasinghe. *ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2022
- OOPSLA 21 Copy-and-Patch Compilation. Haoran Xu and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 5, Issue OOPSLA (Distinguished Paper Award)*, 2021
- OOPSLA 21 Compilation of Sparse Array Programming Models. Rawn Henry, Olivia Hsu, Rohan Yadav, Stephen Chou, Kunle Olukotun, Saman Amarasinghe, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 5, Issue OOPSLA*, 2021
- OOPSLA 20 A Sparse Iteration Space Transformation Framework for Sparse Tensor Algebra. Ryan Senanayake, Changwan Hong, Ziheng Wang, Amalee Wilson, Stephen Chou, Shoaib Kamil, Saman Amarasinghe, and Fredrik Kjolstad. *Proceedings of the ACM on Programming Languages, Volume 4, Issue OOPSLA*, 2020
- DAC 20 Rick Bahr, Clark Barrett, Nikhil Bhagdikar, Alex Carsello, Ross Daly, Caleb Donovan, David Durst, Kayvon Fatahalian, Kathleen Feng, Pat Hanrahan, Teguh Hofstee, Mark Horowitz, Dillon Huff, Fredrik Kjolstad, Taeyoung Kong, Qiaoyi Liu, Makai Mann, Jackson Melchert, Ankita Nayak, Aina Niemetz, Gedeon Nyengele, Priyanka Raina, Stephen Richardson, Raj Setaluri, Jeff Setter, Kavya Sreedhar, Maxwell Strange, James Thomas, Christopher Torng, Leonard Truong, Nestan Tsiskaridze, and Keyi Zhang. Creating an Agile Hardware Design Flow. *Design Automation Conference*, 2020
- PLDI 20 Stephen Chou, Fredrik Kjolstad, and Saman Amarasinghe. Automatic Generation of Efficient Sparse Tensor Format Conversion Routines. *ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2020
- CGO 19 Fredrik Kjolstad, Peter Ahrens, Shoaib Kamil, and Saman Amarasinghe. Sparse Tensor Algebra Optimization with Workspaces. *International Symposium on Code Generation and Optimization*, 2019
- OOPSLA 18 Stephen Chou, Fredrik Kjolstad, and Saman Amarasinghe. Format Abstraction for Sparse Tensor Algebra Compilers. *Proceedings of the ACM on Programming Languages, Volume 2, Issue OOPSLA*, 2018
- OOPSLA 17 Fredrik Kjolstad, Shoaib Kamil, Stephen Chou, David Lugato, and Saman Amarasinghe. The Tensor Algebra Compiler. *Proceedings of the ACM on Programming Languages, Volume 1, Issue OOPSLA (Distinguished Paper Award)*, 2017
- TOG 16 Fredrik Kjolstad, Shoaib Kamil Jonathan Ragan-Kelley, David I.W. Levin, Shinjiro Sueda, Desai Chen, Etienne Vouga, Danny M. Kaufman, Gurtej Kanwar, Wojciech Matusik, and Saman Amarasinghe. Simit: A Language for Physical Simulation. *ACM Transactions on Graphics (TOG, presented at SIGGRAPH)*, 2016

EuroMPI 13 Timo Schneider, Fredrik Kjolstad, and Torsten Hoefler. MPI Datatype Processing using Runtime Compilation. *The 20th European MPI Users' Group Meeting (EuroMPI)* (**Best Paper Award**), 2013

ICSE 11 Fredrik Kjolstad, Danny Dig, Gabriel Acevedo, and Marc Snir. Transformation for Class Immutability. *33rd International Conference on Software Engineering*, 2011

Short Papers and Workshop Publications

PAW-ATM KDRSolvers: Scalable, Flexible, Task-Oriented Krylov Solvers. Proceedings of the Workshop on the Parallel Applications Workshop. David Zhang, Rohan Yadav, Alex Aiken, Fredrik Kjolstad, Sean Treichler. *Proceedings of the Workshop on the Parallel Applications Workshop, Alternatives to MPI+X*, 2025

SPAA Brief 20 Suzanne Mueller, Peter Ahrens, Stephen Chou, Fredrik Kjolstad, and Saman Amarasinghe. Sparse Tensor Transpositions. *ACM Symposium on Parallelism in Algorithms and Architectures (SPAA brief announcement)*, 2020

AVANCÉES 18 David Lugato, Fredrik Kjolstad, Stephen Chou, Saman Amarasinghe, and Shoaib Kamil. Taco: compilation et génération de code d'expressions tensorielles. *AVANCÉES No. 12*, 2018

ASE Tools 17 Fredrik Kjolstad, Stephen Chou, David Lugato, Shoaib Kamil, and Saman Amarasinghe. 32th IEEE/ACM International Conference on Automated Software Engineering. *taco: A Tool to Generate Tensor Algebra Kernels*, 2017

TOG Perspective 16 Gilbert Bernstein, and Fredrik Kjolstad. ACM Transactions on Graphics. *Why New Programming Languages for Simulation?*, 2016

PPoPP Short 12 Fredrik Kjolstad, Torsten Hoefler, and Marc Snir. Automatic Datatype Generation and Optimization. *17th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP short paper)*, 2012

CAP 10 Fredrik Kjolstad, Danny Dig, and Marc Snir. Bringing the HPC Programmer's IDE into the 21st Century through Refactoring. *SPLASH 2010 Workshop on Concurrency for the Application Programmer*, 2010

PPP 10 Fredrik Kjolstad and Marc Snir. Ghost Cell Pattern. *2nd Annual Workshop on Parallel Programming Patterns*, 2010

Awards

- 2026 Distinguished Paper Award, PLDI
- 2025 Google ML and Systems Junior Faculty Award
- 2024 Stanford Tau Beta Pi Teaching Honor Roll
- 2024 Distinguished Artifact Award, ISCA
- 2023 Distinguished Paper Award, PLDI
- 2022 NSF CAREER Award
- 2021 Distinguished Paper Award, OOPSLA
- 2021 Google Research Scholar
- 2020 First Place MIT EECS George M. Sprowls PhD Thesis Award in Computer Science
- 2020 Robert N. Noyce Faculty Fellow, Stanford University School of Engineering
- 2017 Distinguished Paper Award, OOPSLA.
- 2016 Adobe Fellowship.

- 2013 Best Paper Award, EuroMPI
- 2011 Spring Grad Expo Best Poster Award from the University of Illinois at Urbana-Champaign
- 2006 Rosing Prize for best national IT-related student work in 2005 and 2006 from the Norwegian Computer Society
- 2005 Eureka Award for best bachelor project from the Norwegian University of Science and Technology in Gjøvik

Teaching

- 2020–present **CS343d Domain-Specific Programming Models and Compilers**, *Stanford University*.
- 2020–present **CS143 Compilers**, *Stanford University*.

Service

- Program Committees
 - ASPLOS 2027 Program Committee
 - MLSys 2026 Program Committee
 - OOPSLA 2026 Review Committee
 - ASPLOS 2026 Program Committee
 - CACHP 2026 Program Committee
 - Hot Chips 2025 Program Committee
 - PLDI 2025 Program Committee
 - POPL 2025 Program Committee
 - ASPLOS 2025 Program Committee
 - CGO 2025 Program Committee
 - ASPLOS 2024 Program Committee
 - OOPSLA 2024 Review Committee
 - GPCE 2024 Program Committee
 - CC 2024 Program Committee
 - PLDI 2023 Program Committee
 - PPOPP 2023 Program Committee
 - IPDPS 2023 Program Committee
 - CGO 2022 Program Committee
 - PPOPP 2021 Program Committee
 - PLDI 2021 Program Committee
 - SPLASH 2021 Student Research Competition Committee
 - PPOPP 2020 Program Committee
 - CGO 2020 Program Committee
 - PLDI 2020 Student Research Competition Committee
 - PLDI 2020 External Review Committee
 - ASPLOS 2020 External Review Committees
- Chairs
 - PLDI 2025** Sponsorship co-chair
 - PLDI 2024** Sponsorship co-chair
 - PPoPP 2023** Publications chair

Journal and External Reviews	ACM TACO 2019, 2021, 2022, 2024
	ACM TOMS 2021
	ACM TOPC 2020
	ACM TOG 2017
	ACM SIGGRAPH 2019, 2020, 2024, 2026
	ACM SIGGRAPH Asia 2023, 2024
	IEEE Computer Architecture Letters 2022, 2023
	IEEE TPDS 2017, 2021
	NeurIPS 2023
	Springer JPDC 2022
Organizer	2024 Sammaniversary Workshop to celebrate the 60th birthday of Saman Amarasinghe. Hosted at MIT.
	2024 Sparse Workshop (formerly CTSTA) Workshop that brings together leading researchers on programming systems for sparse tensor algebra, graph algorithms, and relational algebra. Colocated with PLDI.
	2023 Workshop on Compiler Techniques for Sparse Tensor Algebra Workshop that brought together leading researchers on sparse tensor algebra compilation and computing. Colocated with PLDI.
	2019 Invited Workshop on Compiler Techniques for Sparse Tensor Algebra Invited workshop that brought together leading researchers on sparse tensor algebra compilation and computing from 11 universities, 6 companies and 3 national labs.
	2012–2013 MIT Programming Languages Offsite Retreat Attended by seven CSAIL professors and their research groups. Re-organized program around many short talks, hosted panels, invited external speakers, and gave opening remarks.
Panels	NSF Review Panel (2021,2022)
	PLDI PL Mentoring Workshop (June, 2020)

Invited Talks

January 2026	Portable Compilation for Future AI, <i>MLPerf-Bench Workshop</i> . Sydney, Australia
November 2025	Domain-Specific Software, Hardware, and their Composition, <i>UCSD Seminar</i> . La Jolla, CA
November 2025	Domain-Specific Software, Hardware, and their Composition, <i>Georgia Tech Guest Lecture</i> . Virtual
October 2025	Domain-Specific Software, Hardware, and their Composition, <i>LLNL Seminar</i> . Livermore, CA
September 2025	Portable Compilation in for Future AI, <i>Google Deepmind Seminar</i> . Mountain View, CA
September 2025	Portable Compilation for Sparse Computation, <i>Apple Seminar</i> . Cupertino, CA
August 2025	AI Kernel Programming Tutorial Introduction, <i>HotChips 2025</i> . Stanford, CA
August 2025	Portable Compilation in an Accelerated World, <i>Dagstuhl Workshop on Software Performance Engineering</i> . Schloss Dagstuhl, Germany
March 2025	Domain-Specific Software for ML, Hardware, and their Composition, <i>Conference on Parsimony and Learning</i> Stanford, CA
Feb. 2025	Domain-Specific Software, Hardware, and their Composition, <i>University of Illinois at Urbana-Champaign, Siebel School Colloquium Series</i> . Urbana, IL.
Jan. 2025	Four Languages for Portability, <i>Case Western Reserve University Lecture</i> . Virtual.
Oct. 2024	Four Languages for Portability, <i>UCLA PL Day</i> . Los Angeles, CA
Oct. 2024	Four Languages for Portability, <i>IBM</i> . Virtual.

Oct. 2024 Portable Compilation for Sparse Computation, *Qualcomm*. Virtual.

Sep. 2024 The Future of Sparse Computing is Compilers, *IEEE HPEC GraphBLAS BoF Keynote*. Virtual.

Sep. 2024 Four Languages for Portability, *KTH Digital Futures Hub*. Stockholm, Sweden.

Sep. 2024 Four Languages for Portability, *Forum on Specification and Design Languages (FDL) Keynote*. Stockholm, Sweden.

July. 2024 Hardware/Software Codesign for Sparse Neural Networks, *Dagstuhl Workshop on Resource-Efficient ML*. Schloss Dagstuhl, Germany.

Jun. 2024 Four Languages for Portability, *Sparse Workshop*. Copenhagen, Denmark.

Apr. 2024 Domain-Specific Software, Hardware, and their Composition, *UC Berkeley Slice Lab*. Berkeley, CA.

Oct. 2023 Hardware/Software Codesign For Sparse Neural Networks, *Meta AI & Systems Co-design Faculty Summit*. Menlo Park, CA.

Sep. 2023 Software and Hardware for Sparse ML. *CODAI Workshop*, Virtual Talk.

Jun. 2023 Portable Compilation for Sparse Computation. *PLDI DRAGSTERS Workshop Keynote*, Orlando, FL.

Jun. 2023 Software and Hardware for Sparse ML. *PLDI CTSTA Workshop*, Orlando, FL.

Apr. 2023 Portable Compilation for Sparse Computation. *Rice Future of Large-Scale Machine Learning Workshop Invited Talk*, Houston, TX.

Nov. 2022 Sparse Tensor Algebra Compilation Tutorial. *NVIDIA*, Virtual Event.

Oct. 2022 Portable Compilation for Sparse Computation. *UC Berkeley Sky Seminar*, Berkeley, CA.

Oct. 2022 Software and Hardware for Sparse ML. *Meta AI HW/SW Codesign Team*, Menlo Park, CA.

Oct. 2022 Portable Compilation of Sparse Computation. *LCPC Invited Talk*, Chicago, IL.

Aug. 2022 Software and Hardware for Sparse ML. *MLSys Chips & Compilers Symposium*, San Jose, CA.

Jun. 2022 Why you would want a compiler in your sparse library. *NVIDIA*, Virtual Event.

Apr. 2022 Sparse Tensor Algebra Compilation. *Meta PyTorch Team*, Menlo Park, CA.

Oct. 2021 Compiling Sparse Array Programming Languages. *Industry-Academia Partnership MIT/Harvard Cloud Workshop*, Virtual Event

Sep. 2021 Sparse Tensor Algebra Compilation. *Amazon Labs*, Cupertino, CA.

Jul. 2021 Fast Compilation and Sparse Compilation. *Google Brain Hardware*, Virtual Event.

Jun. 2021 Sparse Compilation and Fast Compilation. *Intel Research Overview*, Virtual Event.

Apr. 2021 Sparse Tensor Algebra Compilation. *Mathworks*, Virtual Event.

Mar. 2021 Sparse Tensor Algebra Compilation and Sparse Computation. *Accenture Labs Distinguished Researcher Talk*, Virtual Event.

Jan. 2020 Sparse Tensor Algebra Compilation. *SIAM Conference on Parallel Processing for Scientific Computing, invited talk*, Seattle, WA.

Jan. 2020 Sparse Tensor Algebra Optimization. *SIAM Conference on Parallel Processing for Scientific Computing, invited talk*, Seattle, WA.

Dec. 2019 Tensor Algebra Compilation with Workspaces. *Google Compilers and ML Reading Group*, Virtual Event.

Oct. 2019 Optimization in the Sparse Tensor Algebra Compiler. *LCPC Invited Talk*, Atlanta, GA.

Aug. 2019 The Sparse Tensor Algebra Compiler. *Norwegian University of Science and Technology AI Seminar*, Trondheim, Norway.

Jul. 2019 The Sparse Tensor Algebra Compiler. *MIT Fast Code Seminar*, Cambridge, MA

Jun. 2019 Abstraction without Friction. *NSF Workshop on Future Directions for Parallel and Distributed Computing*, Phoenix, AZ

- May 2019 The Sparse Tensor Algebra Compiler. *MIT Graphics Group Seminar*, Cambridge, MA
- Apr. 2019 The Sparse Tensor Algebra Compiler. *Cornell CS, Invited Seminar*, Ithaca, NY.
- Apr. 2019 The Sparse Tensor Algebra Compiler. *Stanford CS, Invited Seminar*, Stanford, CA.
- Mar. 2019 The Sparse Tensor Algebra Compiler. *Georgia Tech CSE, Invited Seminar*, Atlanta, GA.
- Feb. 2019 The Sparse Tensor Algebra Compiler. *Innovations in Software Engineering Conference, Invited Talk*, Pune, India
- Jan. 2019 The Sparse Tensor Algebra Compiler. *UT Austin ECE*, Austin, TX.
- Sep. 2018 The Tensor Algebra Compiler. *University of Texas*, Austin, TX.
- Jun. 2018 The Sparse Tensor Algebra Compiler. *Adobe Research*, Seattle, WA.
- Jun. 2018 The Sparse Tensor Algebra Compiler. *NVIDIA*, Redmond, WA.
- Jun. 2018 The Sparse Tensor Algebra Compiler. *University of Washington*, Seattle, WA.
- Jun. 2018 The Sparse Tensor Algebra Compiler. *Facebook AI*, Menlo Park, CA.
- May 2018 The Sparse Tensor Algebra Compiler. *Stanford*, Stanford, CA.
- May 2018 The Sparse Tensor Algebra Compiler. *UC Berkeley*, Stanford, CA.
- May 2018 The Sparse Tensor Algebra Compiler. *Google Brain*, Mountain View, CA.
- Apr. 2018 The Tensor Algebra Compiler. *Industry-Academia Partnership MIT Cloud Workshop*, Cambridge, MA.
- Mar. 2018 The Tensor Algebra Compiler. *SIAM Conference on Parallel Processing for Scientific Computing*, Tokyo, Japan
- Nov. 2017 The Tensor Algebra Compiler and Simit. *University of Illinois*, Urbana-Champaign, IL.
- Oct. 2017 The Tensor Algebra Compiler. *Microsoft Research*, Redmond, WA.
- Sep. 2016 Simit: A Language for Computing on Sparse Systems. *Intel Research*, Hudson, MA.
- Aug. 2016 Simit: A Language for Computing on Sparse Systems. *Microsoft Research*, Redmond, WA.
- Oct. 2015 Simit: A Language for Computing on Sparse Systems. *MIT Computer Graphics Group Annual Retreat*, Beverly, MA.
- Oct. 2008 Performance Optimization of Embedded 3D Graphics Applications. *ARM Developer's Conference*, Santa Clara, CA.

PhD Students

James Dong
 Christophe Gyurgyik
 Scott Kovach
 Rubens Lacouture (co-advised with Kunle Olukotun)
 Alexander Root
 Shiv Sundram
 Haoran Xu
 Rohan Yadav (co-advised with Alex Aiken)
 Bobby Yan

Alumni

Olivia Hsu (co-advised with Kunle Olukotun), faculty at CMU.
 Trevor Gale (co-advised with Matei Zaharia), research Scientist at Google Deepmind.

Work Experience

Feb 2026 to current	Gimlet Labs , <i>Member of Technical Staff</i>
October 2024 to December 2025	Google Deepmind , <i>Visiting Researcher</i>
November 2022 to February 2024	Amazon , <i>Amazon Visiting Academic</i>
January 2007 to August 2009	ARM Ltd. , <i>Graphics Software Engineer</i>
August 2006 to December 2006	Accenture Technology Solutions , <i>Programmer</i>